

Microprocesadores

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HW # 2

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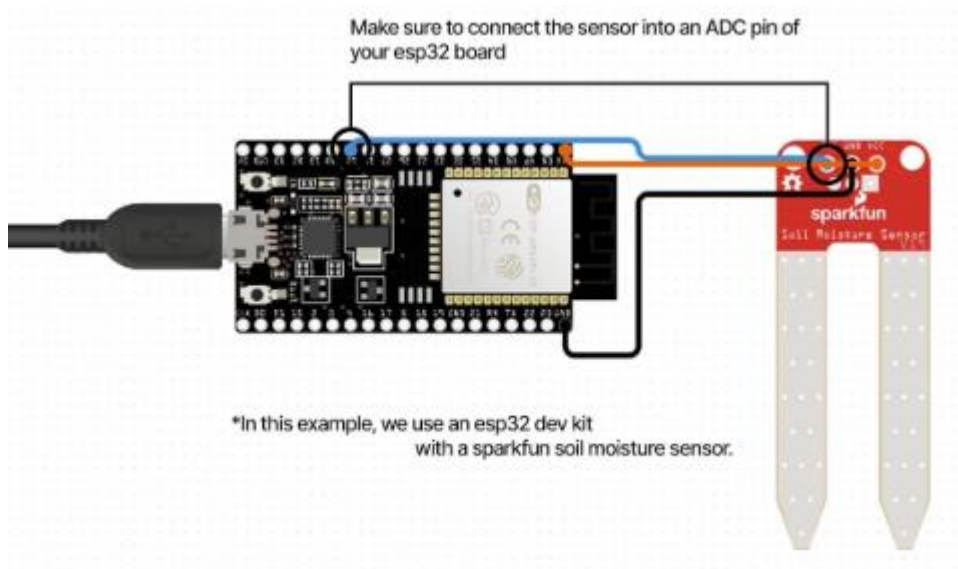
Homework Assignment – ESP32 ADC Single-Shot Moisture Sensor Calibration

Objective

The purpose of this assignment is to design, implement, and calibrate a low-cost soil/sand moisture sensor using the ESP32's ADC (Analog-to-Digital Converter) in single-shot mode. Students will gain practical experience in hardware interfacing, data collection, calibration, and the integration of Python data science tools with embedded C code.

a) Hardware setup

- Two metallic nails were used as electrodes and inserted into a container filled with sand.
- One nail was connected to **3.3 V** and the other to an **ESP32 ADC pin (GPIO34)** through a **voltage divider**, in order to scale the measured voltage to the proper range of the ESP32's ADC (0–3.3 V).
- A common ground (GND) connection was established between the ESP32 and the container.



b) ESP32 configuration

- The project was based on the official `adc/oneshot_read` example from ESP-IDF.
- The code was adapted to perform periodic measurements (1 sample per second) and display both the raw ADC value and the estimated voltage in the serial monitor.
- ADC calibration was enabled through `adc_cali_raw_to_voltage()` to convert raw readings into millivolts.

c) Data acquisition

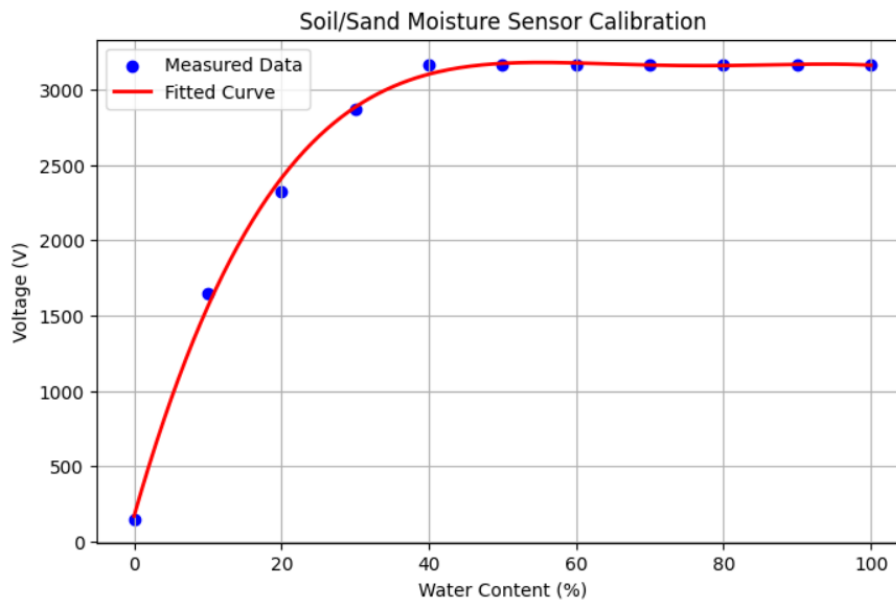
- Experiments were performed with different proportions of water and sand (from completely dry to fully saturated).
- For each condition (% water and % sand), both raw ADC values and corresponding voltages were recorded.
- Data was stored in a .csv file with at least 10 calibration points.

```
I (659299) MoistureCalibration: ADC Raw: 4095 Voltage: 3165 mV
I (660299) MoistureCalibration: ADC Raw: 4095 Voltage: 3165 mV
I (661299) MoistureCalibration: ADC Raw: 4095 Voltage: 3165 mV
I (662299) MoistureCalibration: ADC Raw: 4095 Voltage: 3165 mV
I (663299) MoistureCalibration: ADC Raw: 4095 Voltage: 3165 mV
I (664299) MoistureCalibration: ADC Raw: 4095 Voltage: 3165 mV
I (665299) MoistureCalibration: ADC Raw: 4095 Voltage: 3165 mV
I (666299) MoistureCalibration: ADC Raw: 4095 Voltage: 3165 mV
I (667299) MoistureCalibration: ADC Raw: 0 Voltage: 142 mV
I (668299) MoistureCalibration: ADC Raw: 0 Voltage: 142 mV
I (669299) MoistureCalibration: ADC Raw: 0 Voltage: 142 mV
I (670299) MoistureCalibration: ADC Raw: 0 Voltage: 142 mV
```

Muestra	%Sand	%Water	Raw	mV
1	100	0	0	143
2	90	10	2139	1646
3	80	20	3025	2327
4	70	30	3709	2867
5	60	40	4095	3165
6	50	50	4095	3165
7	40	60	4095	3165
8	30	70	4095	3165
9	20	80	4095	3165
10	10	90	4095	3165
11	0	100	4095	3165

d) Data processing in Python

- The collected data was analyzed using Python libraries such as pandas, numpy, and scikit-learn.
- Different regression models were tested (linear, polynomial, exponential).
- The observed behavior matched best with a **logarithmic curve**, which represents how conductivity increases rapidly with the first increments of water and then stabilizes as saturation is reached. (we use a polynomial form to express this in the data code)



Polynomial coefficients: [0.00000000e+00 1.71881818e+02 -3.58779721e+00 3.24428905e-02
-1.07575758e-04]

Discussion of Sensor Behavior

The calibration results show that the soil/sand moisture sensor exhibits a **nonlinear relationship** between water content (%) and the voltage output measured by the ESP32's ADC. A **third-degree polynomial regression model (cubic)** was used to represent this behavior.

The cubic polynomial provided an excellent fit to the experimental data, capturing subtle variations that a second-degree model could not fully explain. The coefficient of determination (R^2) confirmed that the third-degree model closely follows the measured values, offering a more accurate representation of the sensor's nonlinear response.

The behavior observed can be explained by the physics of water infiltration and conductivity in sand:

- At low water contents, the addition of small amounts of water significantly increases conductivity, leading to noticeable voltage changes.
- At intermediate levels, the response becomes more complex, with the cubic model capturing this curvature more effectively than a simple quadratic.
- At high water contents, the sensor response tends to stabilize, reflecting saturation of the medium and reduced sensitivity to further additions of water.

Conclusion

In this work, a low-cost soil/sand moisture sensor was successfully calibrated using voltage measurements obtained from an ESP32's ADC. The experimental data demonstrated a clear **nonlinear relationship** between water content and sensor voltage, which was effectively modeled using a **third-degree polynomial regression**.

The cubic polynomial provided a highly accurate calibration curve across the full range of water contents, capturing the sensor's varying sensitivity at low, intermediate, and high moisture levels. This calibration allows for reliable estimation of water content from voltage readings, enabling practical applications in soil monitoring and moisture control systems.

Overall, the combination of **experimental data collection, Python-based analysis, and polynomial modeling** offers a robust, cost-effective approach to sensor calibration, and the resulting model can be directly implemented in microcontroller systems like the ESP32 for real-time moisture measurement.